glm-5.1-hf-model-card

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language:

- en
- zh library_name: transformers license: mit pipeline_tag: text-generation

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GLM-5.1



👋 Join our WeChat or Discord community.
📖 Check out the GLM-5.1 blog and GLM-5 Technical report.
📍 Use GLM-5.1 API services on Z.ai API Platform.
➡️ GLM-5.1 will be available on chat.z.ai in the coming days.

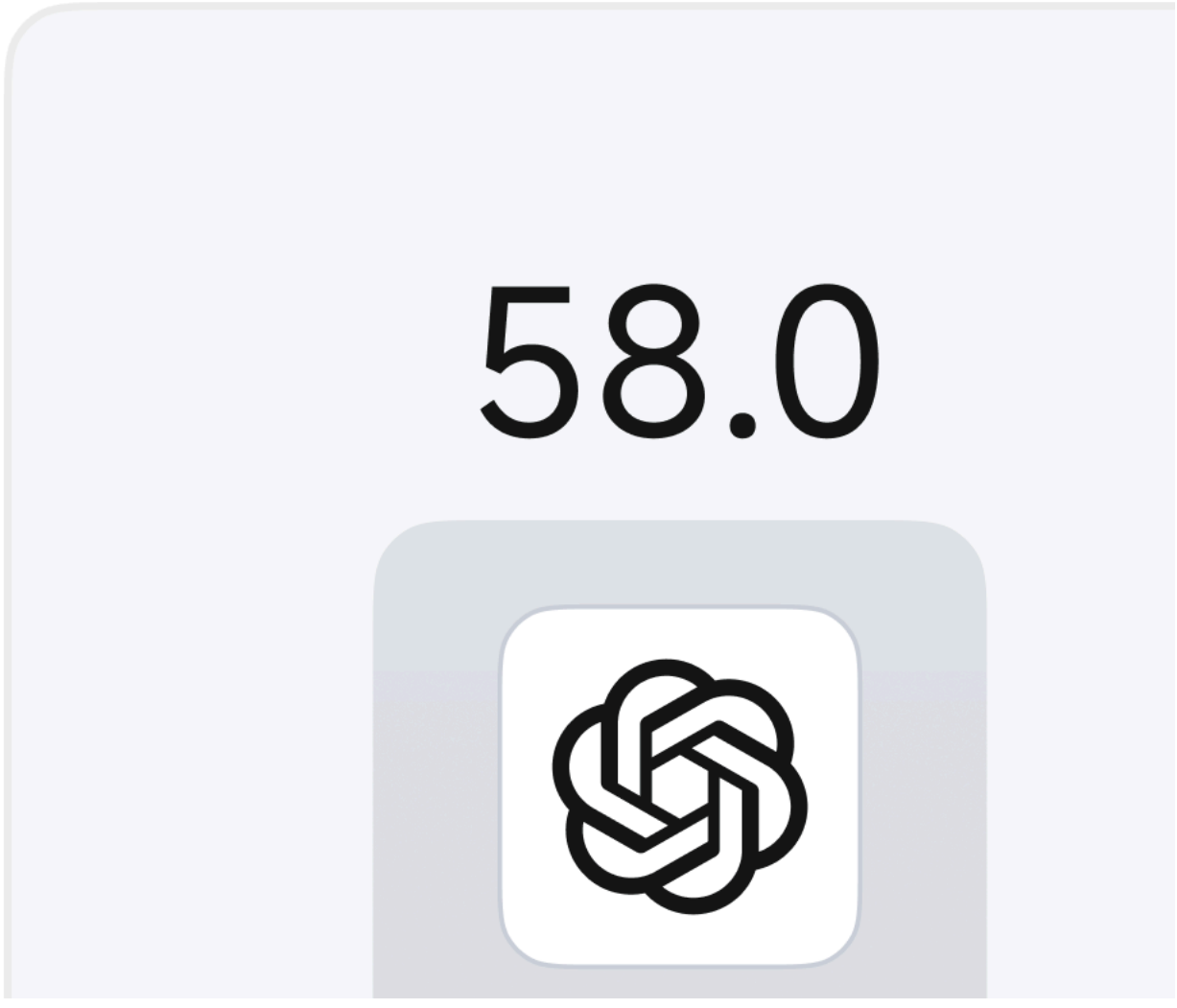
[Paper] [GitHub]

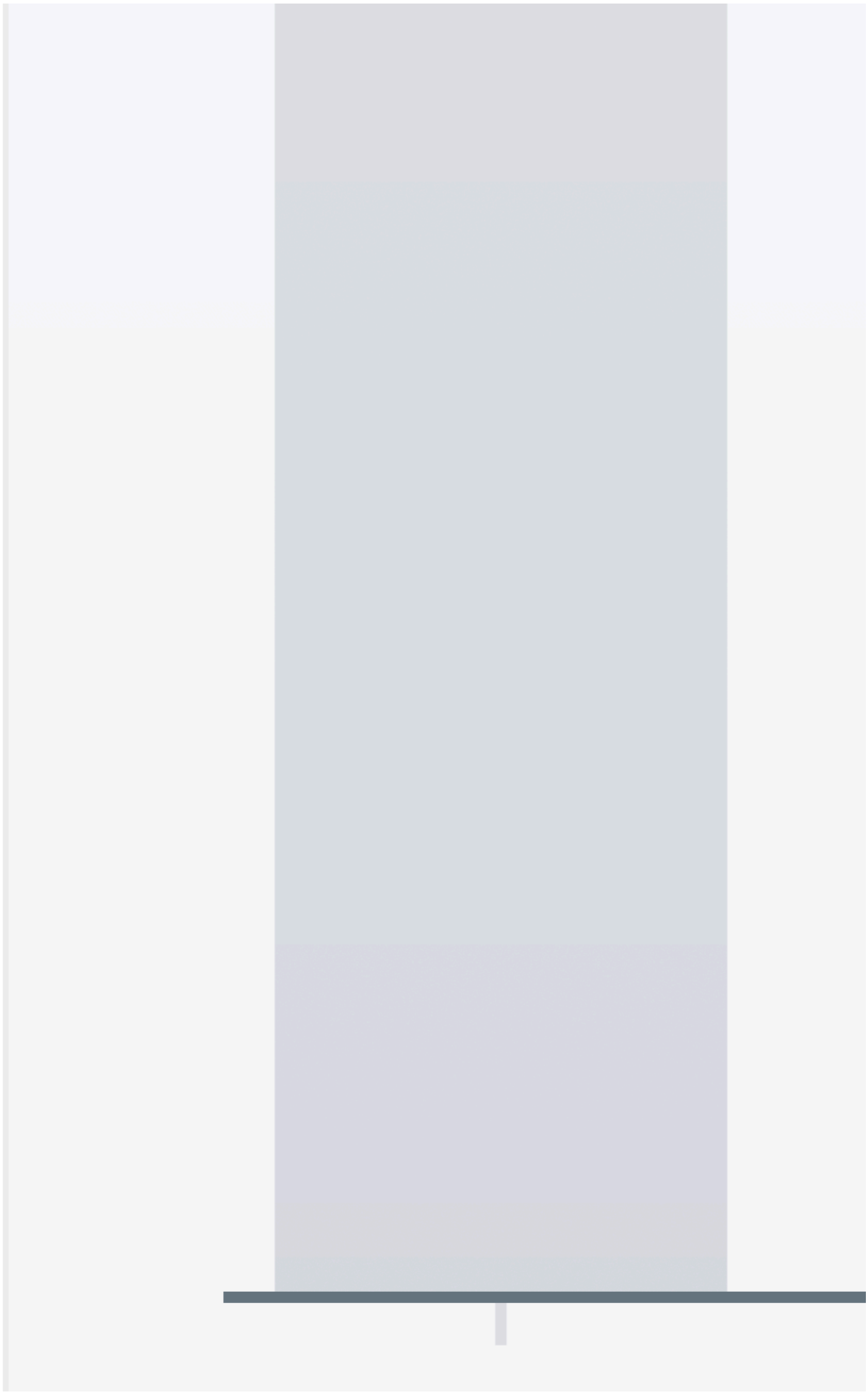
Introduction

GLM-5.1 is our next-generation flagship model for agentic engineering, with significantly stronger coding capabilities than its predecessor. It achieves state-of-the-art performance on SWE-Bench Pro and leads GLM-5 by a wide margin on NL2Repo (repo generation) and Terminal-Bench 2.0 (real-world terminal tasks).

Coding Pe

3 Benchmarks: S







But the most meaningful leap goes beyond first-pass performance. Previous models—including GLM-5—tend to exhaust their repertoire early: they apply familiar techniques for quick initial gains, then plateau. Giving them more time doesn't help.

GLM-5.1, by contrast, is built to stay effective on agentic tasks over much longer horizons. We've found that the model handles ambiguous problems with better judgment and stays productive over longer sessions. It breaks complex problems down, runs experiments, reads results, and identifies blockers with real precision. By revisiting its reasoning and revising its strategy through repeated iteration, GLM-5.1 sustains optimization over hundreds of rounds and thousands of tool calls. The longer it runs, the better the result.

Benchmark

	GLM-5.1	GLM-5	Qwen3.6-Plus	Minimax M2.7	DeepSeek-V3.2	Kimi K2.5	Claude Opus 4.6	Gemini 3.1 Pro	GPT-5.4
HLE	31.0	30.5	28.8	28.0	25.1	31.5	36.7	**45.0**	39.8
HLE (w/ Tools)	52.3	50.4	50.6	-	40.8	51.8	**53.1***	51.4*	52.1*
AIME 2026	95.3	95.4	95.1	89.8	95.1	94.5	95.6	98.2	**98.7**
HMMT Nov. 2025	94.0	**96.9**	94.6	81.0	90.2	91.1	96.3	94.8	95.8
HMMT Feb. 2026	82.6	82.8	87.8	72.7	79.9	81.3	84.3	87.3	**91.8**
IMOAnswerBench	83.8	82.5	83.8	66.3	78.3	81.8	75.3	81.0	**91.4**
GPQA-Diamond	86.2	86.0	90.4	87.0	82.4	87.6	91.3	**94.3**	92.0
SWE-Bench Pro	**58.4**	55.1	56.6	56.2	-	53.8	57.3	54.2	57.7
NL2Repo	42.7	35.9	37.9	39.8	-	32.0	**49.8**	33.4	41.3
Terminal-Bench 2.0 (Terminus-2)	63.5	56.2	61.6	-	39.3	50.8	65.4	**68.5**	-
Terminal-Bench 2.0 (Best self-reported)	69.0 (Claude Code)	56.2 (Claude Code)	-	57.0 (Claude Code)	46.4 (Claude Code)	-	-	-	**75.1** (Codex)
CyberGym	**68.7**	48.3	-	-	17.3	41.3	66.6	38.8	66.3
BrowseComp	**68.0**	62.0	-	-	51.4	60.6	-	-	-
BrowseComp (w/ Context Manage)	79.3	75.9	-	-	67.6	74.9	84.0	**85.9**	82.7
τ^3-Bench	70.6	69.2	70.7	67.6	69.2	66.0	72.4	67.1	**72.9**
MCP-Atlas (Public Set)	71.8	69.2	**74.1**	48.8	62.2	63.8	73.8	69.2	67.2
Tool-Decathlon	40.7	38.0	39.8	46.3	35.2	27.8	47.2	48.8	**54.6**
Vending Bench 2	$5,634.41	$4,432.12	$5,114.87	-	$1,034.00	$1,198.46	**$8,017.59**	$911.21	$6,144.18

Serve GLM-5.1 Locally

The following open-source frameworks support local deployment of GLM-5.1:

- SGLang (v0.5.10+) — see cookbook
- vLLM (v0.19.0+) — see recipes
- xLLM (v0.8.0+) — see example
- Transformers (v0.5.3+) — see transformers docs
- KTransformers (v0.5.3+) — see tutorial

Citation

If you find GLM-5.1 or GLM-5 useful in your research, please cite our technical report:

```bibtex
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      author={GLM-5-Team and : and Aohan Zeng and Xin Lv and Zhenyu Hou and Zhengxiao Du and Qinkai Zheng and Bin Chen and Da Yin and Chendi
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